Semantic Consistency Interaction With Calibration Loss for Remote Sensing Image–Text Retrieval

Jinlong Xu, Yun Ge, Yan Zeng, and Huyang Liu

***Abstract*—Remote sensing image–text retrieval (RSITR) plays a crucial role in the remote sensing (RS) community but faces difficulties due to the heterogeneity between the image and text modalities. While existing methods primarily focus on enhancing cross-modal alignment, they suffer from two key issues. One is that fine-grained local alignment approaches extract local features lacking contextual integration, which leads to semantic inconsistency. The other is that conventional losses fail to adapt to the varying model states of different subtasks, which may lead to the model calibration problems. This implies a mismatch between the confidence scores of the network and the true probabilities. To address these challenges, this study proposes two complementary components. One is a local semantic consistency interaction (LSCI) module, which uses cross-attention and self-attention along with an image–text alignment head to optimize multimodal representations and mitigate context-induced semantic inconsistency. The other is task-specific calibration loss (TC), which dynamically adjusts the gamma parameter in focal or inv-focal loss, allowing the model to prioritize both hard and easy sample pairs according to the calibration error specific to each subtask. The gamma parameter is adjusted by leveraging the previous states of the model and the knowledge of calibration performance on the validation set. By integrating the LSCI module and TC loss, the semantic inconsistency between image and text can be improved, and model calibration can be enhanced. Comparative assessments conducted on three public datasets show that our methods can achieve competitive performance on the RSITR task compared to many existing methods. The source code is available at https://github.com/StrongerPeople/LSCI-TC**

***Index Terms*—Local semantic consistency interaction (LSCI), modality interaction, remote sensing image–text retrieval (RSITR), task-specific calibration loss (TC).**

I. Introduction

With the rapid growth of multimodal remote sensing (RS) data, abundant geographical, environmental, and natural resource information is provided. This information holds significant value for research and applications in military, agriculture, forestry, and marine sciences [1], [2], [3]. However, efficiently extracting rich information from RS data remains a crucial research challenge [4], [5], [6], [7]. To tackle this challenge, vision–language research, including image scene classification [8], image captioning [9], visual question answering [10], text-guided visual grounding [11], image retrieval [12], and image–text retrieval [13], has gained significant attention in this field. Among these tasks, image–text retrieval plays a unique role in cross-modal alignment by bridging visual and linguistic information. Specifically, image–text retrieval involves finding an image that matches a given text query, or conversely, retrieving a text that accurately describes the content of an image. Nevertheless, the complexity of RS images, characterized by nonprominent targets and intricate backgrounds, poses significant challenges to image–text retrieval methods. Furthermore, the necessity of efficiently processing and analyzing vast volumes of RS data has further heightened the demand for dedicated remote sensing image–text retrieval (RSITR) approaches [14], [15], [16], [17].

A key challenge addressed by RSITR is the vision–language heterogeneity gap—fundamental differences between visual and linguistic modalities in data structure, feature distribution, and semantic expression that hinder direct cross-modal interaction and alignment [14], [18], [19]. For instance, RS images have high-dimensional and spatially structured information, containing redundant background noise while requiring preservation of spatial locality; in contrast, text descriptions are discrete symbolic sequences that rely on contextual semantics [20]. To bridge this gap, current RSITR research is mainly divided into two technical types: embedding-based methods and caption-based methods [15]. Embedding-based methods tackle the heterogeneity by mapping image–text features onto a unified embedding space, allowing for direct similarity assessment. This supports highly efficient cross-modal image–text retrieval operations, including text-to-image or image-to-text matching. Moreover, recent transformer-based feature interaction designs have further enhanced the accuracy of cross-modal alignment for these methods. However, a notable drawback is the increased computational cost and parameter count. The need for feature mapping and space construction often leads to heavier model overhead. In contrast, caption-based methods tackle the gap by generating descriptive text for RS images (or retrieving images via text). Their key strength is that strong semantics are directly understandable to humans, which is critical for scenarios like RS image content summarization. However, these methods face a major limitation, that is, the risk of

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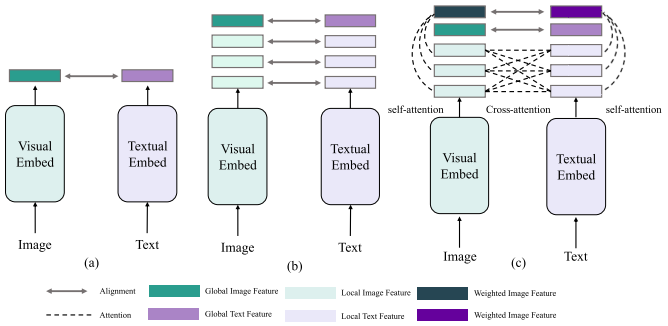


Fig. 1. Visualization of various existing RSITR approaches. (a) Coarse-grained global alignment. (b) Global–local fine-tuning matching methods. (c) Our LSCI method considers both global feature alignment and local weighted feature alignment, thereby achieves better alignment of global image and text embeddings.

model performance collapse [21]. Specifically, when synthetic data are used for training, the model may fail to learn valid patterns, ultimately resulting in such performance degradation.

Combined with the characteristics of RSITR—such as the need for efficient matching of large-scale RS image datasets and strict requirements for spatial feature alignment—the embedding-based methods are more suitable as the focus of research [1], [14]. Their direct similarity measurement capability meets the efficiency demand of cross-modal retrieval, and the improved alignment accuracy via transformer designs can better preserve the spatial locality of RS images. In contrast, the performance collapse risk of caption-based methods is more difficult to mitigate in RS data scenarios, making them less applicable for core retrieval tasks. Therefore, this article will devote particular attention to embedding-based methods.

Existing embedding-based RSITR methods are divided into two directions. The first is global semantic alignment [7], [22], [23], which focuses on mapping image and text features to a unified embedding space for overall semantic matching. It effectively avoids local detail overfitting and ensures basic cross-modal consistency, but it has a notable limitation—it tends to overlook fine-grained semantic correlations between local regions of images and specific words in texts, leading to insufficient alignment precision in scenarios requiring detailed feature matching. The second is global–local fine-tuning matching methods [24], [25], which integrate global semantic constraints with local feature optimization. On one hand, they maintain global embedding alignment, and on the other hand, they fine-tune matches between local image patches and text tokens to enhance fine-grained interaction. However, this direction still faces challenges. It often struggles to balance the weight of global and local information—either over-relying on global semantics to weaken local detail capture, or overemphasizing local features to cause fragmented global alignment. Additionally, its local fine-tuning process lacks adaptability, making it difficult to dynamically adjust based on semantic variations in RS data. To address these issues, improved cross-attention fusion is a promising solution, as shown in Fig. 1. It dynamically focuses on key features, enables bidirectional adaptive interaction between visual and linguistic features, captures fine-grained alignment details while preserving global semantic consistency, and enhances the accuracy and robustness of RSITR.

In addition, existing RSITR methods commonly overlook the calibration error. Traditional retrieval metrics only assess whether the model retrieves semantically matching image–text pairs, but not if the model's prediction confidence aligns with the actual correctness. This discrepancy poses risks for RSITR, which often features scenes with ambiguous targets or noisy backgrounds, and this in turn leads to "calibration problems," specifically high confidence in incorrect retrievals [26], [27]. To address such issues, an increasing number of researchers have started incorporating calibration metrics into both classification and retrieval tasks, aiming to develop more stable and dependable models [26], [27], [28], [29]. For instance, probVLM [27] leverages probabilistic uncertainty scores to evaluate model calibration. Beyond this method, researchers have also proposed standardized calibration metrics to quantify this alignment systematically. A typical example is expected calibration error (ECE) [28], which quantifies this misalignment by binning predictions based on confidence, and subsequently calculating the average discrepancy between predicted confidence and actual accuracy for each bin. Integrating ECE-based calibration loss into training forces the model to adjust its confidence output. This leads to confidence scores that more accurately reflect the true probabilities of the predictions. Consequently, the generalization performance of the model is affected [29]. As shown in Fig. 2, training RSITMD with CLIP using either InfoNCE loss [30] or triplet loss [31] fails to address the model's calibration problems, increasing ECE. Focal loss, as proposed in [32], primarily addresses the imbalance issue in RSITR. Specifically, it downweights the loss of easy-to-match samples while intensifying the loss focus on hard-to-match ones. Notably, this approach has demonstrated potential in reducing calibration error [29], [33]. However, it demands manual tuning of the optimal gamma value, thereby restricting its general applicability and ease of use.

To address these challenges, we initialize our model with contrastive language–image pretraining (CLIP) and integrate it with two proposed components: the local semantic consistency interaction (LSCI) module and the task-specific calibration loss (TC). This integration enhances the model's cross-modal alignment capability on RS image–text retrieval datasets, forming our LSCI-TC framework.

Given the complex semantics in RS datasets, our TC loss is designed to tackle model calibration issues—it dynamically adjusts the gamma parameter based on the model's varying training states, allowing adaptation to different subtasks requirements. By incorporating both inverse-focal loss (Inv-Focal) [29], [34] and focal loss (Focal) [32], this loss effectively reduces the calibration error while balancing the model's focus on hard and easy sample pairs, ensuring robust RS retrieval performance.

We conducted extensive experiments to evaluate the effectiveness of the proposed LSCI-TC model on multiple RS

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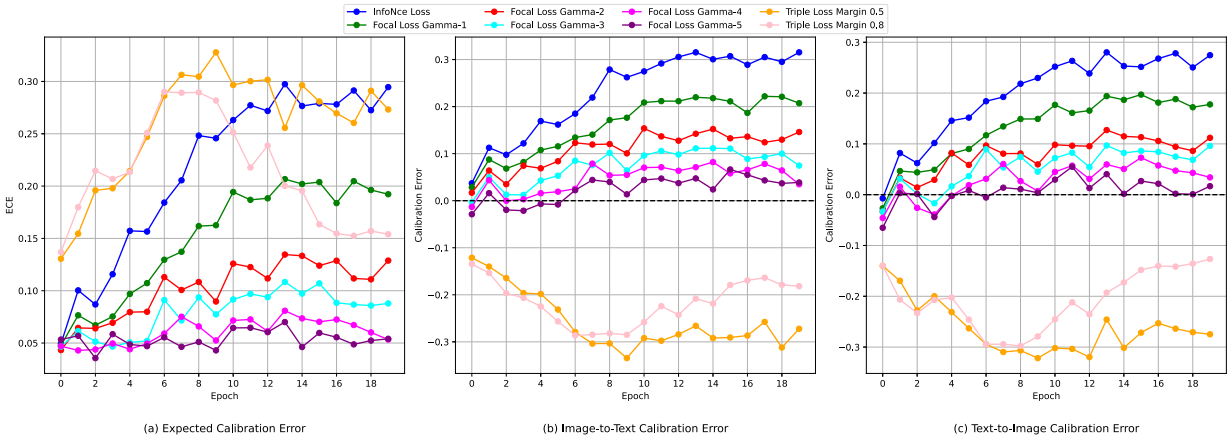


Fig. 2. Calibration behavior of CLIP, trained on RSITMD via InfoNCE loss, focal loss where gamma assumes values in {1, 2, 3, 4, 5}, and triplet loss with margin values in {0.5, 0.8}, was evaluated. These assessments were conducted using 15 equal-mass bins derived from the validation set. (a) Expected calibration error, and (b) and (c) subtasks calibration error. Black dashed horizontal lines in (b) and (c) indicate zero calibration error. These figures demonstrate that while focal loss with gamma = 5 yields the lowest overall calibration error, the optimal gamma differs across subtasks and training epochs.

image–text related datasets. The experimental results demonstrate that the proposed LSCI-TC model outperforms the state-of-the-art methods in terms of retrieval accuracy. The main contributions of our work can be summarized as follows.

1) We propose LSCI-TC, a novel framework that integrates LSCI with a TC, enhancing cross-modal feature integration and model performance optimization while preserving the strengths of the original CLIP architecture.
2) We developed a symmetric LSCI mechanism, which comprises two branches (a text-to-image one and an image-to-text one). These two branches adopt weight sharing to regulate the number of parameters, and the enhanced multimodal features are fused through a multimodal interaction encoder (MIE).
3) We design a TC that integrates both inv-focal loss and focal loss with an adaptive gamma mechanism. This loss dynamically adjusts the focus on hard and easy sample pairs by utilizing the prior states of the model and its calibration performance on the validation set.

II. Related Work

This section begins by reviewing related works in the field of RSITR to further emphasize the research motivation. Subsequently, it explores the advancements in three core components of RSITR models and two mainstream optimization paradigms, followed by an in-depth analysis of existing limitations that motivate our proposed method.

A. RS Image–Text Retrieval

With the advancement of RS technology, image retrieval has achieved significant progress [35], [36], [37]. However, traditional image retrieval methods have limitations when using example images, as the query image alone cannot specify key elements or the exact retrieval object, restricting the versatility of the retrieval system. To address this, RSITR [2], [38] defines retrieval targets semantically, enabling cross-modal searches in RS big data.

A typical RSITR model consists of three main components. There is a visual encoder, a text encoder, and a modality interaction that connects them. Recently, various advances in RSITR have focused on improvements in these components.

Improving the visual encoder aims to better capture multiscale or multilevel image representation [13], [23], [39], [40], [41]. On the one hand, Yuan et al. [15] developed a dynamic fusion module of global and local information. The module effectively combines multilevel visual cues to construct comprehensive feature representations. This enables the enhancement of the capability to capture both semantic and detailed information in RS images. On the other hand, Yuan et al. [39] addressed multiscale scarcity and target redundancy via a multiscale visual self-attention (MVSA), which boosts retrieval performance by extracting multiscale information from RS images and dynamically filtering redundant features. Ma et al. [42] proposed a direction-oriented visual–semantic embedding model, focusing on global semantic mapping to narrow the cross-modal gap.

Some studies have incorporated external knowledge sources into the text encoder to enrich the semantics of the text and to capture concepts and relationships that are not visible in images [22], [43], [44], [45]. Li et al. [43] constructed an RS knowledge graph to support zero-shot RS image scene classification. However, relying solely on the RS knowledge graph is insufficient, as it can only provide domain-specific semantic information and may not capture broader contextual relationships and general semantic aspects of the text. To resolve this limitation, Li et al. [44] utilized a cross-attention mechanism to integrate both domain-specific knowledge [43] and common sense knowledge [46], effectively capturing various semantic aspects of textual descriptions.

Some works mentioned the information asymmetry between image and text, and tried to address this by designing a powerful modality interaction method [25], [47], [48], [49], [50]. Wang et al. [25] designed a multiscale interactive transformer model to explore intermodal relationships and extract multiscale RS image features. Tang et al. [41] proposed a

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BERT-based architecture [51] for fine-grained cross-modal interaction, but its deep transformer encoder results in high computational overhead, making it impractical for large-scale RSITR tasks. Taking a step further, Chen et al. [48] proposed a context-aware local–global semantic alignment (CLGSA) method, whose design is targeted at addressing the inaccuracy of semantic alignment between RS images and texts by integrating both local fine-grained matching and global semantic consistency. Zhou et al. [49] tackled the information asymmetry between RS images and textual data by putting forward a cross-modal retrieval framework that adopts a coarse-to-fine paradigm. In the coarse stage, they adopted a dual-tower architecture to independently extract image–text features, enabling efficient preliminary retrieval via precomputed hash codes. For the fine stage, a single-tower architecture with a joint transformer was designed to fuse image–text features and capture fine-grained cross-modal interactions.

However, in practice, the component-level optimizations discussed earlier remain constrained by two key challenges in RSITR, which are poor adaptability to diverse scenarios and high computational costs for large-scale retrieval. To address these issues, recent research efforts have explored two independent yet complementary technical directions, namely task-adaptive adjustment and sparse representation.

Specifically, to tackle scenario-specific variability, works such as Pan et al. [23], Yang et al. [50], and Huang et al. [52] propose dynamic weighting mechanisms to adjust feature weights or model parameters based on task characteristics. Nevertheless, these methods primarily focus on global adaptation and lack targeted optimization for RS-specific local semantic correlations. This leads to misalignment between local features and cross-modal semantics. On the other front, for balancing retrieval accuracy and computational efficiency, sparse representation techniques have been adopted in works such as Zheng et al. [53] and Zhao et al. [54], which reduce redundant computations by pruning feature dimensions or model structures. While these two technical directions address distinct key challenges in RSITR, they both lack task-specific calibration tailored to the unique semantic gap in RSITR.

This oversight leads to poor generalization across diverse RS scenarios and misaligned cross-modal matching. To address these limitations, we propose an LSCI module and a TC, which are integrated into our framework to capture local semantic correlations and calibrate cross-modal features.

B. Vision–Language Pretrained Model

Due to the inherent correlations between data from different domains, vision–language pretrained (VLP) models based on open domains can better leverage information across various domains, helping the model learn more generalized feature representations. In recent years, VLP models have rapidly developed and achieved encouraging performance in various downstream tasks. They can be divided into three types: single-encoder structures, dual-encoder structures, and hybrid single–dual encoder structures. Single-encoder structures (e.g., SimVLM [55] and ViLBERT [56]) rely on large-scale transformers that have outstanding performance on vision–language understanding tasks. However, single-encoder models can be computationally expensive during image–text retrieval, particularly when employing large-scale modality interaction modules. Conversely, dual-encoder structure models (e.g., GeoRSCLIP [45] and CLIP [57]) reduce computational load by encoding images and texts separately, mapping image and text features into the same semantic space, while demonstrating impressive performance in image–text retrieval tasks. As a complement to the two aforementioned structures, hybrid single–dual encoder structures are designed to balance the performance advantages of single-encoder models and the computational efficiency of dual-encoder models (e.g., ALIGN [58] and BLIP [59]). For RSITR scenarios, which have unique data characteristics, research on optimizing hybrid structures or leveraging their advantages to address domain-specific challenges has also attracted attention. For example, Zhou et al. [49] tackled the information asymmetry between RS images and textual data by introducing a cross-modal retrieval framework that adopts a coarse-to-fine paradigm, with its design subtly aligning with the balance-oriented logic of hybrid single–dual encoder structures.

Given the inherent task gap between natural image–text retrieval and RSITR, applying CLIP, which is not specifically designed for the RS domain, may not perform optimally on RSITR tasks. The characteristics and challenges distinctive of RS data, such as multiple targets, nonprominent targets, complex and redundant backgrounds, and domain-specific semantics, necessitate tailored approaches for effective RSITR. Our LSCI module stands out in managing complex correspondences between RS images and texts.

III. Methodology

The general framework of the proposed LSCI-TC method is illustrated in Fig. 3. A more comprehensive discussion of the components and their interactions can be found in the following subsections, specifically Sections III-A–III-D.

A. Overview of the Proposed Pipeline

The LSCI-TC framework comprises feature extraction, LSCI, and TC. The feature extraction module focuses on extracting local text and visual features, aiming to capture fine-grained semantics from both modalities. Our training strategy for feature extraction involves two TC losses and soft-target labels obtained via self-training—both of which enhance the discriminative ability and alignment capabilities of the generated local features. Then, in the LSCI module, the similarity between the extracted image and text features is calculated, and the top-k similar pairs are selected for modality interaction; the model is further optimized using an image–text matching (ITM) loss. In the following, we describe the details of these modules.

B. Local Semantic Consistency Interaction

The LSCI has three key components: image-to-text multihead cross-attention (ITMCA), text-to-image multihead cross-attention (TIMCA), and MIE. Subsequently, through feature

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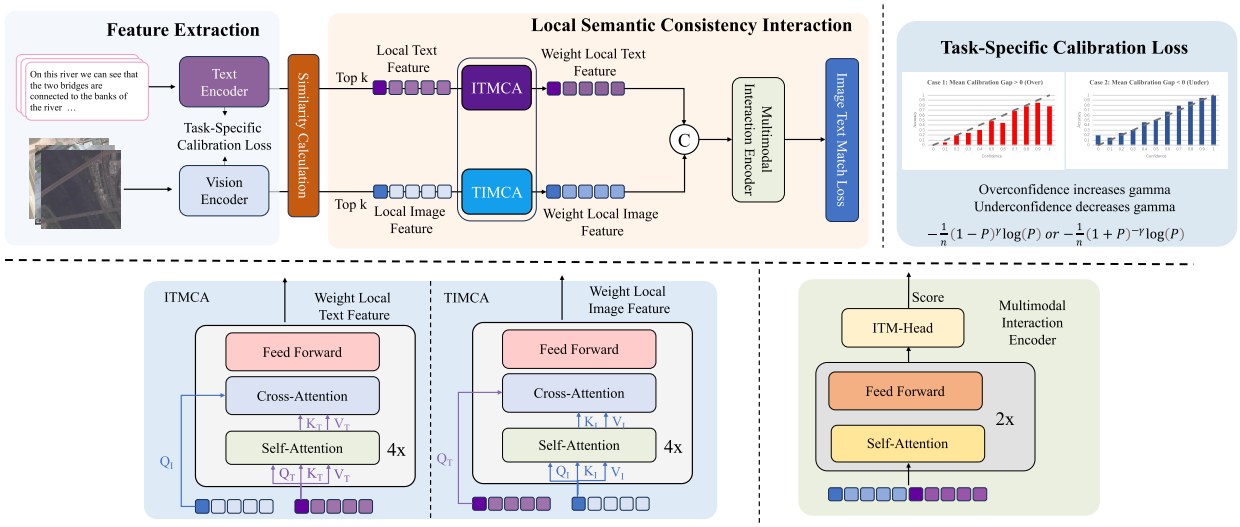


Fig. 3. Overview of the proposed LSCI-TC model is illustrated in the pipeline, which consists of a feature extraction, a local semantic consistency modality interaction module, and a TC mechanism. The model first encodes RS images and texts using separate vision and text encoders to extract discriminative local features. Notably, these encoders are optimized by the TC, which dynamically adjusts their training focus to ensure that the learned local features align better with the demands of RS image–text retrieval. Local features are then enhanced via cross-modal interaction. TIMCA and ITMCA generate weighted local features. These features are further refined by an MIE with stacked self-attention layers, enabling deep fusion.

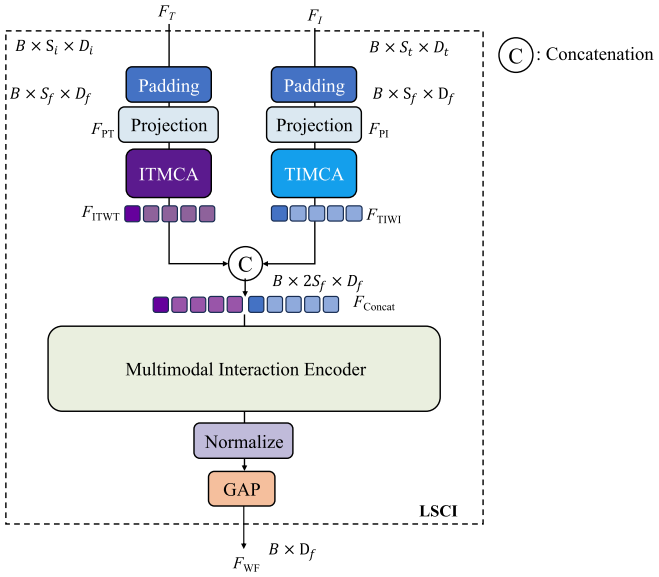


Fig. 4. Pipeline of the LSCI module includes three core components: ITMCA, TIMCA, and MIE. These components collectively enable deep cross-modal interaction between image and text features. Key operations: padding (maintains feature dimensional consistency), projection (linear layer aligning modalities to a shared space), normalize (layer normalization), and GAP.

interaction operations, such as concatenation, the weighted image features are obtained.

As depicted in Fig. 4, the image features F_I and the text features F_T are fed into the ITMCA/TIMCA as inputs.

Since the image features F_I and the text features F_T are not in the same feature dimension, we need to project them into a shared dimensional space. Before projection, padding operations are applied to both image and text features to ensure that they have consistent sequence lengths, which helps align the features for effective projection. The formulas for the projected features are shown in the following equations:

$$F_{\mathrm{PI}} = \mathrm{Projection}\left(\mathrm{Padding}\left(F_I\right)\right) \tag{1}$$

$$F_{\mathrm{PT}} = \mathrm{Projection}\left(\mathrm{Padding}\left(F_T\right)\right) \tag{2}$$

where $F_I \in \mathbb{R}^{B\times S_i\times D_i}$ represents the original image features, with B as the batch size, S_i as the original image sequence length, and D_i as the feature dimension for images. $F_T \in \mathbb{R}^{B\times S_t\times D_t}$ stands for the original text features, where S_t is the original text sequence length and D_t is the feature dimension for text. After padding, both F_I and F_T are adjusted to have a unified sequence length S_f, ensuring they can be effectively projected into the shared space. The projected features $F_{\mathrm{PI}} \in \mathbb{R}^{B\times S_f\times D_f}$ and $F_{\mathrm{PT}} \in \mathbb{R}^{B\times S_f\times D_f}$ are the final outputs, both mapped to a shared feature dimension D_f.

1) Image-to-Text and Text-to-Image Cross-Attention: Taking the image-to-text weighted feature (F_{ITWI}) as an example. We utilize multihead cross-attention mechanisms for cross-modal feature interaction between images and texts. As shown in Fig. 5, for the ith attention head, we linearly project input features into query Q_i, key K_i, and value V_i vectors. The formulas are shown as follows:

$$Q_i = F_{\mathrm{PI}}W_i^Q, \quad K_i = F_{\mathrm{PT}}W_i^K, \quad V_i = F_{\mathrm{PT}}W_i^V \tag{3}$$

here, $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{D_f\times S_f}$ are trainable parameters.

Then, the weights obtained from the text-to-image similarity are employed to weight the image features. This process is shown in (4). The output of the ith head is given by (5)

$$W_{\mathrm{TI}} = \frac{\phi\left(F_{\mathrm{PI}}\right)\phi\left(F_{\mathrm{PT}}\right)^T}{\sqrt{D}} = \frac{Q_iK_i^T}{\sqrt{D}} \tag{4}$$

$$\hat{F}_i^V = \mathrm{softmax}\left(W_{\mathrm{TI}}\right)V_i \quad (i=1,\ldots,k) \tag{5}$$

where $\phi(\cdot)$ denotes L2-normalization along the feature dimension. The weighted image features F_{ITWI} and text features

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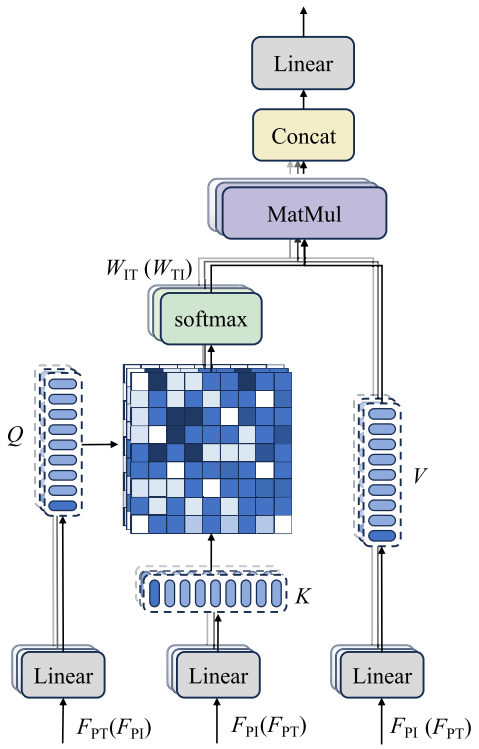


Fig. 5. Architecture of the image-to-text/text-to-image multihead cross-attention (ITMCA/TIMCA) module. The cross-modal attention mechanism computes affinity scores between visual and textual features through scaled dot-product operations. These attention weights are then applied to adaptively highlight image regions semantically relevant to the textual context. "MatMul" denotes matrix multiplication.

F_{TIWI} are computed through multiple heads. The formula is shown as follows:

$$F_{\mathrm{ITWI}} = \mathrm{Concat}\left(\hat{F}_1^V, \ldots, \hat{F}_k^V\right) \times W_0. \tag{6}$$

The multihead output comes into being by concatenating the outputs from all the heads, and then projecting them via a learnable weight matrix W_0. In the process, weights derived from the image-to-text retrieval are employed to enhance text features. A similar approach is used to compute F_{TIWT}, with the input features adjusted accordingly.

2) Multimodal Interaction Encoder: To fuse distinct multimodal features, these features are combined via concatenation, with the corresponding formula presented as follows:

$$F_{\mathrm{concat}} \in \mathbb{R}^{B\times L\times D_f} = \mathrm{Concat}\left(F_{\mathrm{ITWI}}, F_{\mathrm{TIWI}}\right) \tag{7}$$

where B denotes the batch size, L represents the total sequence length of the two concatenated multimodal features ($L = 2S_f$), and D_f stands for the feature dimension.

For cross-modal feature interaction, we adopt a self-attention framework based on mutual information, which takes F_{concat} as the direct input to model the intrinsic correlation between different modal components in the concatenated feature. The calculation process of this self-attention with mutual information awareness is illustrated in (9)

$$Q = F_{\mathrm{concat}} \cdot W_Q, K = F_{\mathrm{concat}} \cdot W_K, V = F_{\mathrm{concat}} \cdot W_V \tag{8}$$

here, $W_Q, W_K, W_V \in \mathbb{R}^{D_f\times S_f}$ are trainable parameters

$$\hat{F}_M = \mathrm{softmax}\left(\frac{QK}{\sqrt{D}}\right) V \tag{9}$$

where $\hat{F}_{\mathrm{M}} \in \mathbb{R}^{B\times S_f\times D_f}$. The concatenation operation preserves the distinct characteristics of each individual feature while combining them into a unified representation. It allows the model to retain the unique information from each feature, which can be crucial for tasks that require complex understanding of different aspects.

Finally, a combination of normalization and global average pooling (GAP) is applied to further integrate the different visual features, aiming to refine the fused representation by reducing distribution discrepancies and aggregating global semantic information. This final integration step is demonstrated in the following equation:

$$F_{\mathrm{WF}} = \mathrm{GAP}\left(\mathrm{Normalize}\left(\hat{F}_M\right)\right). \tag{10}$$

C. Task-Specific Calibration via Soft Labels

Currently, prevalent contrastive learning losses (e.g., InfoNCE loss) struggle to handle complex semantic relationships between image–text pairs in RS data. This is largely due to the prevalence of numerous semantically similar image–text pairs and the lack of a thorough account of the distributional properties of the data, thereby failing to address the model calibration problems, as shown in Fig. 2. We propose the TC, which dynamically adjusts the gamma parameter in the loss function based on the calibration error of each subtask and the model's historical states. This mechanism enables the model to balance its focus on hard and easy sample pairs.

Specifically, given a mini-batch sample, a random selection of N positive RS image–text pairs is made, represented as $\{(I_i, T_i) \mid i = 1, 2, \ldots, N\}$. Corresponding final image and text embeddings are generated for each pair, leading to a set $\{(v_i, t_i) \mid i = 1, 2, \ldots, N\}$. The similarity calculation is performed as $s_{i,j} = \mathrm{sim}(v_i, t_j)$. To determine the probability of match for each pair (i, j), a softmax function with a temperature parameter τ is applied, as illustrated in 11. This probability indicates the likelihood of image and text being correctly matched, taking into account all other pairs in the mini-batch

$$P_{i,j} = \frac{\exp\left(s_{i,j}/\tau\right)}{\sum_{j=1}^{N}\exp\left(s_{i,j}/\tau\right)} = \mathrm{softmax}\left(s_{i,j}\right). \tag{11}$$

Therefore, we can get the image-to-text focal loss and text-to-image focal loss as (12) and (15)

$$L_{\mathrm{vt}}\left(v_i, t_i\right) = -\left(1 - P_{i,i}\right)^{\gamma}\log\left(P_{i,i}\right) \tag{12}$$

$$s'_{i,i} = t_i \times {v_i}^T \tag{13}$$

$$P'_{i,i} = \frac{\exp\left(s'_{i,i}\right)/\tau}{\sum_{j=1}^{N}\exp\left(s'_{i,j}/\tau\right)} \tag{14}$$

$$L_{\mathrm{tv}}\left(t_i, v_i\right) = -\left(1 - P'_{i,i}\right)^{\gamma}\log\left(P'_{i,i}\right). \tag{15}$$

The focal loss, where $\gamma > 0$, is commonly used to address the issue of the model's overconfidence. Additionally, we

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TABLE I
EXPERIMENTAL RESULTS ON THE RSICD DATASET

Method	Pre-training Dataset	Training Dataset	Image/Text Backbone	Image to Text			Text to Image			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
GaLR [15]	———	RSICD	ResNet-18+ppyolo/GRU	6.59	19.85	31.04	4.69	19.48	32.13	18.96
MSIT [25]	———	RSICD	linear-projector/BERT	6.46	23.69	38.45	4.96	20.74	35.86	21.69
SCAT-PRG [65]	———	RSICD	ResNet-50/GRU	10.88	26.32	39.66	8.06	25.43	40.31	25.11
IEFT [41]	———	RSICD	ViT-B-32/BERT	8.11	27.45	43.25	6.26	24.00	39.98	24.84
CLIP-FT [57]	WIT(CLIP)	RSICD	ViT-B-32/transformer	15.37	33.48	47.20	12.86	35.91	53.44	33.04
CUP [66]	WIT(CLIP)	RSICD	ViT-L-14/transformer	12.17	35.36	52.45	13.14	36.14	51.36	33.43
PEFT [67]	WIT(CLIP)	RSICD	ViT-B-32/transformer	14.13	31.51	44.78	11.63	33.92	50.73	31.12
GLISA [47]	WIT(CLIP)	RSICD	———	19.52	40.44	52.28	14.75	39.50	55.46	36.99
RemoteCLIP [22]	RET-3 + DET-10 + SEG-4	———	ViT-B-32/transformer	17.02	37.97	51.51	13.71	37.11	54.25	35.26
RemoteCLIP [22]	RET-3 + DET-10 + SEG-4	———	ViT-L-14/transformer	18.39	37.42	51.05	14.73	39.93	56.58	36.35
GeoRSCLIP [45]	RS5M	RSICD	ViT-B-32/transformer	22.14	40.53	51.78	15.26	40.46	57.79	38.00
GeoRSCLIP [45]	RS5M	RET-2	ViT-B-32/transformer	21.13	41.72	55.63	15.59	41.19	57.99	38.87
HarMA [52]	LAION	RSICD	ViT-B-32/transformer	16.36	34.48	47.74	12.92	37.17	53.07	33.62
HarMA [52]	RS5M+RET-2	RSICD	ViT-B-32/transformer	20.52	41.37	54.66	15.84	41.92	**59.39**	38.95
PIR-CLIP [23]	RS5M	RSICD	ResNet-50+ViT-B-32/transformer	**26.62**	40.71	50.32	**21.35**	43.59	54.27	39.48
PIR-CLIP [23]	RS5M	RET-2	ResNet-50+ViT-B-32/transformer	25.62	43.18	53.06	19.91	42.74	55.15	39.95
LSCI-TC	RS5M	RSICD	ViT-B-32/transformer	20.40	44.55	55.80	17.73	43.22	56.06	39.63
LSCI-TC	RS5M	RET-2	ViT-B-32/transformer	22.59	**46.02**	**62.39**	19.10	**45.48**	57.47	**42.18**

TABLE II
EXPERIMENTAL RESULTS ON THE RSITMD DATASET

Method	Pre-training Dataset	Training Dataset	Image/Text Backbone	Image to Text			Text to Image			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
GaLR [15]	———	RSITMD	ResNet-18+ppyolo/GRU	14.82	31.64	42.48	11.15	36.68	51.68	31.41
MSIT [25]	———	RSITMD	linear-projector/BERT	14.37	33.18	48.74	9.34	37.62	58.81	33.67
SCAT-PRG [65]	———	RSITMD	ResNet-50/GRU	16.50	33.25	48.49	12.55	41.86	61.07	35.62
IEFT [41]	———	RSITMD	ViT-B-32/BERT	15.48	37.61	51.40	11.10	37.99	58.75	35.38
CLIP-FT [57]	WIT(CLIP)	RSITMD	ViT-B-32/transformer	21.46	45.35	57.96	18.58	52.43	70.35	44.35
CUP [66]	WIT(CLIP)	RSITMD	ViT-L-14/transformer	19.28	52.83	71.11	23.23	45.80	60.84	45.61
PEFT [67]	WIT(CLIP)	RSITMD	ViT-B-32/transformer	23.67	44.07	60.36	20.10	50.63	67.97	44.47
GLISA [47]	WIT(CLIP)	RSITMD	———	28.21	50.94	62.50	20.50	54.80	72.19	48.41
RemoteCLIP [22]	RET-3 + DET-10 + SEG-4	———	ViT-B-32/transformer	27.88	50.66	65.71	22.17	56.46	73.41	49.38
RemoteCLIP [22]	RET-3 + DET-10 + SEG-4	———	ViT-L-14/transformer	28.76	52.43	63.94	23.76	59.51	74.73	50.52
GeoRSCLIP [45]	RS5M	RSITMD	ViT-B-32/transformer	30.09	51.55	63.27	23.54	57.52	74.60	50.10
GeoRSCLIP [45]	RS5M	RET-2	ViT-B-32/transformer	32.30	53.32	67.92	25.04	57.88	74.38	51.81
HarMA [52]	LAION	RSITMD	ViT-B-32/transformer	25.81	48.37	60.61	19.92	53.27	71.21	46.53
HarMA [52]	RS5M+RET-2	RSITMD	ViT-B-32/transformer	32.74	53.76	69.25	25.62	57.65	74.60	52.27
PIR-CLIP [23]	RS5M	RSITMD	ResNet-50+ViT-B-32/transformer	30.97	58.19	71.02	25.62	55.27	75.00	52.68
PIR-CLIP [23]	RS5M	RET-2	ResNet-50+ViT-B-32/transformer	**44.25**	65.49	**75.00**	30.13	55.44	**68.54**	**56.70**
LSCI-TC	RS5M	RSITMD	ViT-B-32/transformer	35.39	60.61	72.56	30.26	58.31	63.05	53.37
LSCI-TC	RS5M	RET2	ViT-B-32/transformer	39.15	**65.70**	**75.00**	**34.25**	**61.15**	64.20	56.58

TABLE III
EXPERIMENTAL RESULTS ON THE NWPU-CAPTIONS DATASET

Method	Pre-training Dataset	Training Dataset	Image/Text Backbone	Image to Text			Text to Image			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
SCAN [68]	———	NWPU-Captions	CNN/GRU	12.6	40.1	81.6	9.5	35.7	63.2	40.45
IMRAM [69]	———	NWPU-Captions	Faster R-CNN/GRU	4.3	31.0	65,3	7.1	25.3	51.1	30.68
MTFN [70]	———	NWPU-Captions	CNN/GRU	11.9	37.0	65.3	7.1	25.3	51.1	34.36
PFAN [71]	———	NWPU-Captions	CNN/GRU	11.2	32.6	71.0	7.6	22.9	47.5	32.13
CLIP-FT [57]	WIT(CLIP)	NWPU-Captions	ViT-B-32/transformer	21.01	53.71	71.07	11.57	35.68	51.82	40.82
GeoRSCLIP-FT [45]	RS5M	NWPU-Captions	ViT-B-32/transformer	21.23	54.50	71.58	12.02	36.10	52.24	41.28
LSCI-TC	RS5M	NWPU-Captions	ViT-B-32/transformer	**23.33**	**57.33**	**74.73**	**12.86**	**37.53**	**53.95**	**43.29**

introduce the inverse-focal loss, which is given by the following equation:

$$L_{\mathrm{vt}}(v_i, t_i) = -\left(1 + P_{i,i}\right)^{\gamma} \log\left(P_{i,i}\right) \tag{16}$$

where $\gamma > 0$. This loss function is designed to handle the problem of the model's underconfidence.

As shown in Fig. 6, focal loss and inverse-focal loss assign different weights to hard and easy sample pairs. Specifically, the focusing parameter gamma smoothly adjusts the rate at which easy examples are downweighted. When gamma = 0, focal loss is equivalent to InfoNCE loss, and as gamma is increased the effect of the modulating factor is likewise increased. In contrast, the inverse-focal loss exhibits

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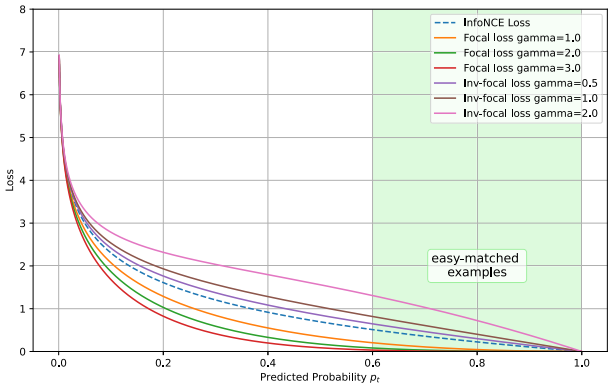


Fig. 6. Focal loss and inv-focal loss curves under different gamma values.

TABLE IV
ABLATION EXPERIMENTS OF THE LSCI-TC MODEL ON RSITMD AND RSICD DATASETS

Dataset	No.	InfoNCE	TC	Soft-label	LSCI	mR(%)	ECE(%)
RSITMD	1	✓				45.56	21.91
	2	✓		✓		46.26	18.22
	3	✓			✓	52.42	16.67
	4	✓		✓	✓	52.85	14.21
	5		✓			45.71	17.68
	6		✓	✓		46.34	16.35
	7		✓		✓	52.36	14.93
	8		✓	✓	✓	**53.37**	**14.15**
RSICD	1	✓				33.04	23.12
	2	✓		✓		34.39	18.33
	3	✓			✓	38.12	21.87
	4	✓		✓	✓	38.85	16.92
	5		✓			33.38	19.95
	6		✓	✓		34.65	17.81
	7		✓		✓	38.37	17.23
	8		✓	✓	✓	**39.63**	**16.78**

the opposite behavior. Extensive experiments, as shown in Table VI, confirm that this adaptive gamma adjustment significantly impacts model performance, enhancing its generalization ability, consistent with conclusions in [29] and [33].

In image–text retrieval task, the training data can be formalized as a set $\{(v_i, t_i, y_i^{\text{true}}) \mid i = 1, 2, \ldots, n\}$. Each triplet consists of an image v_i, the corresponding text description t_i, and a binary label $y_i^{\text{true}} \in \{0, 1\}$ indicating whether the pair is correctly matched. A model is trained to output a matching probability $\hat{P}_{\hat{y}}$ for each input pair, reflecting the confidence of the model in the image–text alignment. For clarity, $\hat{y}$ is defined as the matching prediction of the model, and the probability $\hat{P}_{\hat{y}}$ as the confidence. A model is regarded as perfectly calibrated when the confidence probability $\hat{P}_{\hat{y}}$ aligns with the correct prediction $\hat{y}$ of the model. For example, if the model assigns an average matching score of 0.8 to a set of image–text pairs, 80% of these pairs are expected to be correctly matched.

To quantify calibration, CE and ECE are employed as evaluation metrics for the model. Equal-mass bins [60] are utilized to assess model performance. The approximated formulas are presented in (19) and (20)

$$C_i = \frac{1}{|B_i|} \sum_{j \in B_i} \hat{P}_{\hat{y},j} \tag{17}$$

$$A_i = \frac{1}{|B_i|} \sum_{j \in B_i} \mathbb{F}\left(\hat{y}_j = y_j^{\text{true}}\right) \tag{18}$$

$$\text{CE}_i = C_i - A_i \tag{19}$$

$$\text{CE}_m = \frac{1}{M} \sum_{i=1}^{M} C_i \; - \; A_i \tag{20}$$

where B_i indicates the number of examples in each bin. The function $\mathbb{F}(\hat{y}_j = y_j^{\text{true}})$ produces a set of binary values (0 or 1). M is the number of bins. The calibration error for bin i (denoted as CE_i) is the difference between the confidence C_i and the accuracy A_i in that bin, that is, $C_i - A_i$. The average calibration error across all bins is defined by (20). A positive CE_m indicates the model is overconfident for the subtask, while a negative value suggests underconfidence. Conversely, when the CE_m is less than zero, it suggests that the model is in underconfidence for that subtasks. Based on the CE_m, we have designed a simple adaptive gamma method as follows:

$$\gamma_i \; = \; e^{\text{CE}^{\text{val}}} \times \gamma_{i-1} \tag{21}$$

where CE^{val} represents CE_m on the validation set. For different subtasks, image-to-text obtains $\text{CE}_{\text{vt}}^{\text{val}}$ and text-to-image obtains $\text{CE}_{\text{tv}}^{\text{val}}$, resulting in two distinct calibration errors. γ_{i-1} denotes the previous value of γ. The TC is presented in Algorithm 1.

Subsequently, we offer a detailed explanation of the TC. Since cross-modal RS image retrieval consists of two subtasks, image-to-text retrieval and text-to-image retrieval, we first initialize the gamma values for both subtasks, denoted as $\gamma_{t=0}^{\text{vt}}$ and $\gamma_{t=0}^{\text{tv}}$. Then, at the tth epoch, we evaluate the model's performance on the validation set, obtaining the average calibration error for the subtasks, represented as $\text{CE}_{\text{vt}}^{\text{val}}$ and $\text{CE}_{\text{tv}}^{\text{val}}$, respectively. When the subtask's $\text{CE}_m > 0$, it indicates that the model is in overconfidence for that subtask, and we increase the gamma value to address this overconfidence. When the subtask's $\text{CE}_m < 0$, it signifies that the model is in underconfidence for that subtask, and we decrease the gamma value accordingly.

Of course, this approach has a problem. When the gamma value updates to 0, it stops changing. To address this, we set a switching threshold $S_{\text{th}} = 0.1 \times (\gamma_{t=0}^{\text{vt}} \quad \text{or} \quad \gamma_{t=0}^{\text{tv}})$. When the gamma value decreases or increases to 0.1 times its initial value and further adjustment in the same direction is desired, we use S_{th} to directly switch the sign of the gamma update. This allows for a rapid transition to the alternative loss function to address the model's underconfidence or overconfidence more effectively. To simplify our notation, a negative γ denotes the inverse-focal loss. Thus, $\gamma_t > 0$ means focal loss, while $\gamma_t < 0$ indicates the inv-focal loss.

Finally, due to the inverse-focal loss, when the gamma value is less than 1, the optimization function becomes a nonconvex curve, which may negatively impact the model's performance. Therefore, it is necessary to set a minimum limit for gamma. However, we also do not want the gamma to keep increasing, as this may cause the model to focus only on the hard-to-match sample pairs in the data and ignore the easy-to-match sample pairs. Hence, we set the maximum value ($\gamma_{\max}$) and the minimum value ($\gamma_{\min}$) for the change in gamma.

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TABLE V

RESULTS OF THE ABLATION STUDY ON DIFFERENT COMPONENTS OF THE LSCI MODULE, CONDUCTED WITH THE INFONCE LOSS ON RSITMD DATASET

No.	ITM	TIM	MIE	Image to Text			Text to Image			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
1	✓			30.94	60.38	67.56	27.69	54.51	60.60	50.28
2		✓		31.04	61.50	68.90	25.83	52.71	57.50	49.58
3	✓		✓	32.38	62.04	70.11	25.10	53.26	61.17	50.67
4		✓	✓	33.70	61.69	70.22	24.25	52.41	60.46	50.45
5	✓	✓	✓	34.58	63.57	70.66	25.22	56.43	64.11	52.42

TABLE VI

IMPACT OF INITIAL GAMMA VALUES ON MODEL PERFORMANCE (MR AND ECE) ON THE RSITMD DATASET FOR TC LOSS AND FOCAL LOSS VARIANTS

Configuration		Image to Text			Text to Image			Metrics	
Loss	Init Gamma	R@1	R@5	R@10	R@1	R@5	R@10	mR	ECE
Focal or Inv-Focal	-2	30.75	61.96	72.37	25.86	57.26	62.97	51.86	23.51
	-1	31.20	61.33	70.76	28.34	54.99	63.35	51.66	18.40
	1	33.53	62.33	71.36	25.10	57.32	63.18	52.14	16.98
	2	32.72	61.04	71.05	27.70	55.36	62.89	51.79	16.48
Sigmoid Focal	1	25.88	53.98	65.26	26.63	57.25	66.54	49.26	18.40
	2	29.42	53.98	67.47	29.15	57.83	64.73	50.43	20.45
Dynamic-Recall Focal	1	31.49	60.22	69.37	28.13	57.54	65.41	52.02	17.76
	2	28.84	57.64	68.73	29.53	56.37	65.78	51.14	16.10
TC	-2	32.75	61.42	71.85	25.62	55.88	64.22	51.96	21.00
	-1	32.82	62.72	72.37	27.42	55.16	61.38	51.98	16.71
	1	34.23	61.81	70.72	26.80	56.37	64.22	52.36	14.93
	2	31.49	61.39	70.99	26.98	55.99	61.73	51.43	15.12
Quality Focal Loss or Inv-Focal + Soft-label	-2	32.50	61.20	71.83	29.10	54.30	65.20	52.35	16.65
	-1	32.21	61.50	71.03	27.84	54.67	65.92	52.17	15.42
	1	32.95	62.84	72.45	28.61	55.67	64.62	52.86	14.58
	2	33.10	61.50	72.13	29.30	55.20	64.28	52.58	14.62
TC + Soft-label	-2	33.95	61.88	72.45	28.98	56.20	65.58	53.26	16.02
	-1	32.54	62.80	72.25	28.90	56.27	64.82	52.93	14.81
	1	33.58	62.88	73.45	28.98	56.67	64.62	53.37	14.15
	2	32.10	61.84	72.38	28.20	56.17	65.56	52.71	13.64

Therefore, the expression for the TC in the image–text retrieval task is as follows, with $\gamma_t > 0$ as an example:

$$\begin{cases} L_{\text{vt}}(v_i, t_i) = -\left(1 - P_{i,i}\right)^{\gamma_t} \log\left(P_{i,i}\right) \\ \gamma_t = \gamma_{t-1} \times e^{\text{CE}_{\text{vt}}^{\text{val}}} \\ L_{\text{tv}}(t_i, v_i) = -\left(1 - P_{i,i}\right)^{\gamma_t} \log\left(P_{i,i}\right) \\ \gamma_t = \gamma_{t-1} \times e^{\text{CE}_{\text{tv}}^{\text{val}}}. \end{cases} \tag{22}$$

The soft-label module leverages the intramodal similarity scores derived from self-training, along with weighted one-hot labels, to formulate a softened target. This approach helps alleviate the rigid one-to-one constraint within the distribution.

Next, the intramodal self-similarities are used as soft labels to supervise the image-to-text and text-to-image correspondences respectively. In practice, we use the weighting of hard labels and that of soft labels as the final softened targets. This ensures training stability and better generalization.

The soft labels are computed using the following formulas:

$$P_{i,j}(v_i, t_j) = \frac{\exp\left(\text{sim}_{(v_i,t_j)}/\tau\right)}{\sum_{j=1}^{N} \exp\left(\text{sim}_{(v_i,t_j)}/\tau\right)} \tag{23}$$

$$P_{i,j}(t_i, v_j) = \frac{\exp\left(\text{sim}_{(t_i,v_j)}/\tau\right)}{\sum_{j=1}^{N} \exp\left(\text{sim}_{(t_i,v_j)}/\tau\right)}. \tag{24}$$

The soft alignment label is defined as follows:

$$\widetilde{y}_{i,j}^{\text{it}} = \alpha y_{i,j} + (1 - \alpha) P_{i,j}\left(v_i, t_j\right) \tag{25}$$

$$\widetilde{y}_{i,j}^{\text{ti}} = \alpha y_{i,j} + (1 - \alpha) P_{i,j}\left(t_i, j_j\right) \tag{26}$$

where $y_{i,j}$ stands for the one-hot label. $P_{i,j}(v_i, t_j)$ and $P_{i,j}(t_i, v_j)$ represent the image-to-text probability and text-to-image probability, respectively. α is the hyperparameter that controls the degree of smoothing.

Since the traditional focal loss is designed for one-hot labels (with (v_i, t_i) denoting positive pairs only) and is not capable of handling soft labels, we adopt the method from quality focal loss [61]. The improved formula is given in the following equation:

$$L'_{\text{vt}}\left(v_i, t_j\right) = \begin{cases} -\left|\widetilde{y}_{i,j}^{\text{it}} - P_{i,j}\right|^{\gamma_t} \widetilde{y}_{i,j}^{\text{it}} \log\left(P_{i,j}\right), & \gamma_t > 0 \\ -\left|\widetilde{y}_{i,j}^{\text{ti}} + P_{i,j}\right|^{-\gamma_t} \widetilde{y}_{i,j}^{\text{ti}} \log\left(P_{i,j}\right), & \gamma_t \le 0. \end{cases} \tag{27}$$

A similar formulation holds for text-to-image $L'_{\text{tv}}(t_i, v_j)$.

D. Overall Objective Function

The loss function proposed in this study consists of three components: two cross-modal TC loss, denoted as L_{vt} and L_{tv}, and one ITM loss. Specifically, the ITM loss is designed to

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Algorithm 1 Task-Specific Calibration Loss

Require: $D_{\text{train}} = \{\mathbf{I}_i, \mathbf{T}_i\}_{i=1}^{N_{\text{train}}}$, $D_{\text{val}} = \{\mathbf{I}_i, \mathbf{T}_i\}_{i=1}^{N_{\text{val}}}$;

Bin initialization:
  for $i = 1$**to**M **do**
    $B_{t=0,i}$ = Quantile$(i-1, M)$ // Define Equal-mass bins
  end for // End of bin initialization
  $\gamma_{t=0}^{vt} = R_1$ // Initialize $\gamma_{t=0}^{vt}$
  $\gamma_{t=0}^{tv} = R_2$ // Initialize $\gamma_{t=0}^{tv}$
Training: for $t = 0$ to T do
  $Loss_t = 0$;
  for $n = 1$ to N_{train} do
    (p_n^{vt}, p_n^{tv}) = model(I, T) // p_n^{vt} denotes the probability of image to text matching, and p_n^{tv} denotes the probability of text to image matching.
    if $\gamma_t^{vt} \geq 0$ then
      $Loss_t += -(1 - p_n^{vt})^{\gamma_t^{vt}} \log p_n^{vt}$ // Focal loss;
    else if $\gamma_t^{vt} < 0$ then
      $Loss_t += -(1 + p_n^{vt})^{|\gamma_t^{vt}|} \log p_n^{vt}$ // Inverse-Focal loss;
      if $\gamma_t^{tv} \geq 0$ then
      $Loss_t += -(1 - p_n^{tv})^{\gamma_t^{tv}} \log p_n^{tv}$;
    else if $\gamma_t^{tv} < 0$ then
      $Loss_t += -(1 + p_n^{tv})^{|\gamma_t^{tv}|} \log p_n^{tv}$;
    end if
  end for
  γ**-update step:** for i=1 to M do
    Re-compute $C_{\text{val},i}$ and $A_{\text{val},i}$ using D_{val}, re-calculate $\text{CE}_{\text{vt}}^{\text{val}}$ and $\text{CE}_{\text{tv}}^{\text{val}}$;
  end for
  for γ_t **in**$\{\gamma_t^{vt}, \gamma_t^{tv}\}$ **do** // Update subtasks γ
    if $\gamma_t \geq 0$ then
      $\gamma_{t+1} = \min\{\gamma_{\max}, \gamma_t \cdot e^{\text{CE}_{\text{vt}}^{\text{val}}} or \quad \gamma_t \cdot e^{\text{CE}_{\text{tv}}^{\text{val}}}\}$
      if $|\gamma_{t+1}| < S_{\text{th}}$ then
        $\gamma_{t+1} = -S_{\text{th}}$ // Switch to inv-focal;
    end if
    else if $\gamma_t < 0$ then
$\gamma_{t+1} = \max\{\gamma_{\min}, \gamma_t \cdot e^{-\text{CE}_{\text{vt}}^{\text{val}}} or \quad \gamma_t \cdot e^{-\text{CE}_{\text{tv}}^{\text{val}}}\}$
      if $|\gamma_{t+1}| < S_{\text{th}}$ then
      $\gamma_{t+1} = S_{\text{th}}$ // Switch to Focal;
      end if
  end for
end for

optimize the semantic matching between image and text pairs, and its formulation is given by the following equation:

$$L_{\text{itm}} = -\frac{1}{N}\sum_{(i,j)}\left[y_{\text{ij}} \cdot \log \sigma\left(s\left(v_i, t_j\right)\right) + \left(1 - y_{\text{ij}}\right) \cdot \log\left(1 - \sigma\left(s\left(v_i, t_j\right)\right)\right)\right] \tag{28}$$

Here N denotes the total count of image–text pairs in the batch, y_{ij} is a binary label (1 for matched image–text pairs, 0 for mismatched pairs), $s(v_i, t_j)$ denotes the similarity score between the feature of image v_i and the feature of text t_j, and $\sigma(\cdot)$ is the softmax function that maps the similarity score to a probability within the range [0, 1].

The overall objective function of the model is the sum of the three loss components, which is formulated as follows:

$$L_{\text{all}} = L'_{\text{vt}} + L'_{\text{tv}} + L_{\text{itm}}. \tag{29}$$

This integrated loss function jointly optimizes the cross-modal feature alignment (via TC losses) and the fine-grained semantic matching between image and text pairs (via ITM loss), thereby boosting the model's ability to capture cross-modal semantic consistency.

IV. Experiments and Analysis

Within this section, we begin by outlining the experiment settings, which include the dataset, the training samples, evaluation metrics, and the implementation details. Next, we compare the LSCI-TC model with other methods and conduct ablation studies in order to verify the effectiveness of the method we proposed.

A. Datasets

In our experimental evaluations, our method was assessed on three benchmark datasets—RSITMD [13], RSICD [62], and NWPU-Captions [63]—that are commonly used in the RSITR task. Additionally, two frequently utilized datasets, RET-2 [22] and RS5M [45], were also incorporated into our study.

1) *RSICD Dataset:* RSICD [62] serves as an RS image–text dataset that contains 10 921 images. Each image has a resolution of 224 × 224 and five sentences. We follow the dividing in [62], utilizing 7862 images for training, 1966 for validation, and 1093 for testing.
2) *RSITMD Dataset:* RSITMD [13] functions as an RS image–text dataset with 4743 images, each sized 256 × 256 and accompanied by five descriptive sentences. Following the data partition strategy in [13], we use 3435 images for training, 856 for validation, and 452 for testing.
3) *NWPU-Captions Dataset:* NWPU-Captions [63] is an RS image–text dataset comprising 34 650 images (each 256 × 256 in size), with five annotated sentences provided per image. We follow the data splitting method specified in [63], employing 25 200 training images, 3150 validation images, and 3150 testing images.
4) *RS5M Dataset:* RS5M [45] is an RS image–text dataset that includes roughly five million image–text pairs. For our study, the RS5M dataset is employed as the pretraining dataset.
5) *RET-2 Dataset:* RET-2 [22] is formed by combining the RSICD and RSITMD datasets. To avoid data leakage, it undergoes data deduplication—a process aligned with the practices of RemoteCLIP [22] and GeoRSCLIP [45].

B. Evaluation Metrics

To evaluate our LSCI-TC model, we adopt two widely used metrics $R@k$ (where k = 1, 5, 10) and mR [64] as the main evaluation metrics. A relation label of 1 indicates that the RS image and text are a positive sample pair. Otherwise, they are

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considered as a negative sample pair. Initially, one must obtain the k relation labels corresponding to the top-k matching scores between the query and all retrieval entities. $R@k$ represents the probability of retrieving the associated text for a given image or the associated image for a given text. Therefore, $R@k$ can be defined as follows:

$$R@k = \frac{1}{N}\sum_{i=1}^{N} \text{Score}\,(i,k) \tag{30}$$

where N means the volume of the query set, i means the ith query set, and Score(i,k) denotes whether the positive items corresponding to the ith query appear in the top-k retrieval samples or not. If the positive sample exists in the top-k retrieval results, Score$(i,k) = 1$. Otherwise, Score$(i,k) = 0$. Furthermore, mR is the average of all R@k.

In addition to R@k and mR, we also incorporate the ECE metric to evaluate the calibration of our model. The ECE is a measure of the difference between the model's confidence in predictions and the actual accuracy of those predictions. It is defined as follows:

$$\text{ECE} = \frac{1}{2}\left(\underbrace{\sum_{i=1}^{M}\frac{1}{M}\left|\text{conf}\,(B_i) - \text{acc}\,(B_i)\right|}_{\text{Image Retrieval}} + \underbrace{\sum_{j=1}^{M}\frac{1}{M}\left|\text{conf}\,(B_j) - \text{acc}\,(B_j)\right|}_{\text{Text Retrieval}}\right) \tag{31}$$

where conf(B_i) represents the model's confidence in its predictions for the ith bin, while acc(B_i) denotes the actual accuracy of the predictions within that bin. M is the total number of bins, and n is the number of instances within each bin. In image–text retrieval, the two subtasks are distinguished by indices i and j. Within this context, i corresponds to the ECE for image-to-text retrieval, and j corresponds to the ECE for text-to-image retrieval. The ECE is the mean of the subtasks' ECE values, providing a more comprehensive assessment of the calibration performance of the model.

C. Implementation Details

Our model training adopts a two-phase paradigm: a pretraining phase and a fine-tuning phase. Specifically, pretraining is performed in parallel on 2 NVIDIA RTX A40 GPUs using the RS5M dataset, with a batch size of 512, a total of 20 pretraining epochs, a learning rate that varies from $1e^{-5}$ to $1e^{-7}$, and automatic mixed-precision (AMP) employed to optimize training efficiency. While the training phase is conducted on a single NVIDIA 3090 GPU equipped with 24 GB memory, and a learning rate that varies from $2e^{-5}$ to $2e^{-7}$. The LSCI-TC framework, the core of our model, comprises an image encoder module, a text encoder module, and an LSCI module. Among these components, the LSCI module consists of two cross-attention mechanisms, each configured with eight attention heads, and the MIE is set to have eight self-attention heads.

During training, we use data augmentation that follows the approach described in [52] and [55]. For the TC, the initial gamma values for image-to-text retrieval, text-to-image retrieval, and ITM are all set to 1.0, the smooth label α is set to 0.8, and the top-k parameter is configured to 40.

D. Comparison With State-of-the-Art Methods

In this section, we conducted a comprehensive comparative analysis of our LSCI-TC model with traditional image–text retrieval methods and VLP methods.

1) Traditional Methods: Some methods focus on extracting local information to enhance the fine-grained perception between images and text. For example, GaLR [15] employs an attention-based multilevel module to effectively integrate global and local features. SCAT-PRG [65] improves local feature extraction within salient objects through a spatial-channel attention mechanism. Both methods focus on emphasizing key regions and features, thereby improving retrieval accuracy. Recent methods focus on enhancing discriminative feature representations and semantic alignment for visual–language methods. MSIT [25] achieves this through a parameter-efficient multiscale hierarchical learning architecture, where discriminative features are refined via progressive cross-scale information aggregation. IEFT [41], on the other hand, employs an information interacting enhancement module to capture cross-modal relationships and improve visual feature discriminability by exploring information from the local.

2) VLP Fine-Tuning Methods: These methods, which pretrain on larger scale datasets, can be fine-tuned and adapted to suit RSITR tasks. Some focus on efficiently transferring visual–language models from natural to RS domains. For example, CUP [66] employs a prompt generation module to create RS-oriented prompt tokens, capturing object-level RS image details. PEFT [67], using a lightweight adapter module, fine-tunes the pretrained model while keeping most parameters frozen. Both methods maintain the pretrained model's strong generalization ability and integrate specific RS knowledge, significantly improving RS image–text retrieval performance with parameter efficiency. Other methods focus on improving feature representation and alignment quality in cross-modal matching. For instance, MSA-CLIP [24] enhances cross-modal feature learning by leveraging cross-attention for multiscale alignment, strengthening the model's ability to capture detailed interactions between visual and textual modalities.

As evidenced by the quantitative comparisons in Tables I–IV, our LSCI-TC surpasses existing methods across all metrics in this experiment. For image-to-text retrieval, LSCI-TC achieves R@1, R@5, and R@10 values of 22.59%, 46.02%, and 62.39%, respectively, on the RSICD (RET-2) dataset. For text-to-image retrieval, the corresponding metrics are 19.10%, 45.48%, and 57.47%, with a mean recall (mR) of 42.18%. Compared to the traditional method SCAT-PRG [65], our method shows enhancements of 11.71%, 19.70%, and 22.73% for R@1, R@5, and R@10 in image-to-text retrieval, and 11.04%, 20.05%, and 17.16% in text-to-image retrieval. Against the VLP method GeoRSCLIP [45] (RSICD

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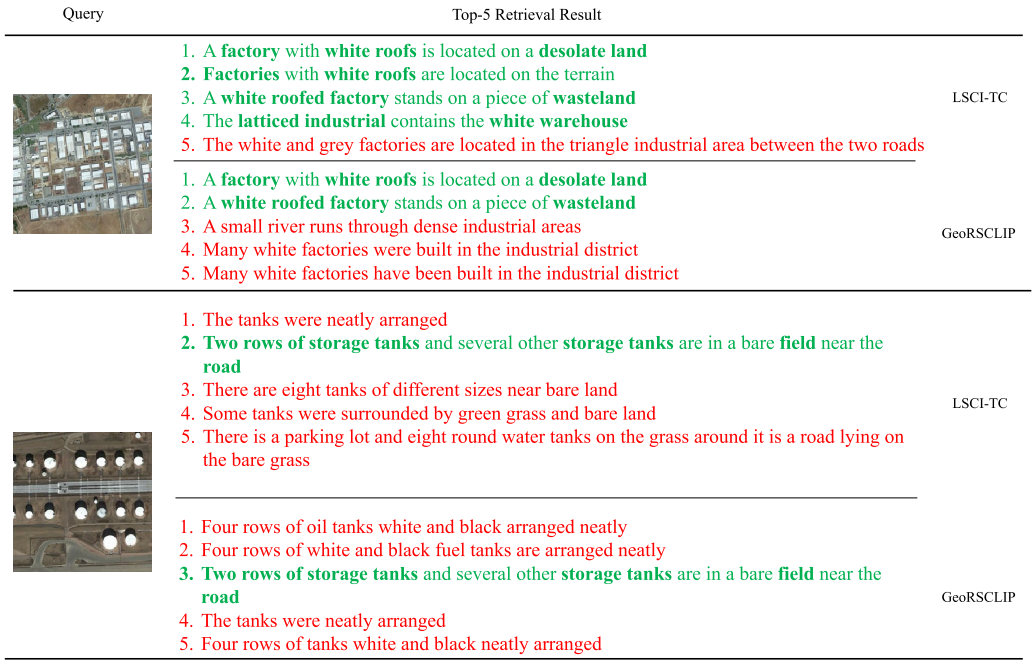


Fig. 7. Top-5 image-to-text retrieval results obtained from the RSITMD test dataset. Matching and mismatched results are highlighted in green and red, respectively.

RET-2), there is a 3.31% improvement in mR, reaching 42.18% compared to its 38.87%. The strong performance on RSICD underscores the effectiveness of our LSCI module in capturing latent cross-modal correlations. On the RSITMD (RET-2) dataset, our method also achieves leading results. For image-to-text retrieval, LSCI-TC reaches R@1, R@5, and R@10 of 39.15%, 65.70%, and 75.00%, while text-to-image retrieval yields 34.25%, 61.15%, and 64.20% (mR = 56.58%). Compared to the previous representative work MSIT [25], our LSCI-TC exhibits considerable performance improvements: 24.78%, 32.52%, and 26.26% increases in R@1, R@5, and R@10 for image-to-text retrieval, and 24.91%, 23.53%, and 5.39% for text-to-image retrieval. Even against the SOTA PIR-CLIP [23], LSCI-TC leads in text-to-image R@1/R@5 (34.25%/61.15% versus 30.13%/55.44%) despite using a lighter ViT-B-32 backbone. These outcomes highlight the generalization ability of LSCI-TC. In summary, our LSCI-TC consistently achieves the best performance across all metrics on RSICD, RSITMD, and NWPU-Captions benchmarks, demonstrating robustness and broad applicability for RSITR.

E. Visualization of Results

For a visual illustration of our proposed model's effectiveness, we present three types of qualitative experimental results, including image-to-text retrieval visualization, text-to-image retrieval visualization, and word-level localization visualization.

1) Image-to-Text Retrieval: In Fig. 7, we present the top-5 image-to-text retrieval results on the RSITMD dataset. The first example shows that GeoRSCLIP is easily distracted by irrelevant elements, such as the mention of a "small river." In contrast, the proposed LSCI-TC model demonstrates a more comprehensive and accurate understanding of the image–text relationship.

2) Text-to-Image Retrieval: In Fig. 8, the top-5 text-to-image retrieval results on RSITMD are presented. The focus of GeoRSCLIP solely on "pond" while neglecting "two" leads to a mismatch. However, LSCI-TC can accurately find the most similar image among various options through textual descriptions.

3) Word-Level Localization: Grad-CAM [72] is used to visualize word-level localization in images for image–text pairs. As shown in Fig. 8, LSCI-TC demonstrates more precise responses to specific nouns than GeoRSCLIP, accurately identifying the corresponding regions. The examples indicate that visual–spatial attention is allocated to semantically relevant areas in the image, with visually relevant regions receiving higher attention weights.

F. Ablation Studies

In this section, we conduct extensive ablation studies on RSITMD, RSICD, and NWPU-Caption to analyze the impact of these modules.

1) Ablation Experiments on the Components of LSCI-TC: First, we design four sets of experiments to combine these modules and optimize them with different loss functions on RSITMD.

1) *No. 1:* The model is optimized solely with the InfoNCE loss, serving as the baseline method without any proposed modules.
2) *No. 2:* The model is optimized with the InfoNCE loss together with the soft-label strategy.
3) *No. 3:* The model is enhanced with the LSCI module and optimized using the InfoNCE loss.

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Fig. 8. (a) Text-to-image retrieval examples on RSITMD test dataset. The top-5 retrieved images are shown from rank1 to rank5. (b) Grad-CAM heatmaps for locating the correspondence between words and image regions.

4) *No. 4:* The model includes both the LSCI module and the soft-label strategy, and is optimized with the InfoNCE loss.
5) *No. 5:* The CLIP model is optimized solely using the task-specific calibration (TC) loss without the LSCI module.
6) *No. 6:* The model incorporates the soft-label strategy and is trained with the TC loss, without using the LSCI module.
7) *No. 7:* The model employs the LSCI module and is optimized using the TC loss.
8) *No. 8:* The complete LSCI-TC model, which integrates the LSCI module, the soft-label strategy, and the TC loss.

Table IV displays the results of our ablation studies. In the following content, we provide a detailed comparison and analysis of these results to clarify how each component impacts the overall performance of the model.

1) *Contribution of the LSCI Module:* The integration of the LSCI module (No. 3) leads to a substantial improvement over the baseline (No. 1), with an mR increase of 6.86% and an ECE reduction of 5.24% on RSITMD set. This result highlights the limitation of conventional feature extraction methods that process modalities in isolation. The LSCI module explicitly models cross-modal interactions at the feature level, enabling the model to capture fine-grained semantic alignments prior to similarity computation.
2) *Effect of the Soft-Label Strategy:* Incorporating the soft-label strategy with the LSCI module and InfoNCE loss (No. 4) yields further improvements, increasing mR by 0.43% and reducing ECE by 2.46% compared to the configuration without it (No. 3) on RSITMD set. Unlike hard labels that enforce strict binary assignments, soft labels act as a form of semantic smoothing, reflecting the continuous and often ambiguous nature of cross-modal relevance. This result is consistent with the work of [59], [73], and [74] collectively underscoring the importance of soft semantic alignment in overcoming the inherent challenges of cross-modal retrieval.
3) *Synergy Between TC Loss and LSCI Module:* Equipping the TC loss with the LSCI module (No. 7) results in a notable performance boost compared to using TC loss alone (No. 5), with a 6.64% increase in mR and a 2.75% reduction in ECE on RSITMD set. This synergy underscores a key insight: adaptive loss functions like TC depends heavily on the quality of input features. While TC loss adjusts the gamma based on prediction confidence, its effectiveness is maximized when features are already well-aligned through cross-modal interaction. This suggests that simply replacing loss functions is insufficient [74]. Instead, codesigning feature interaction and adaptive optimization is critical for achieving well-calibrated and accurate retrieval.
4) *Supplementary Analysis on TC Loss Versus InfoNCE:* Notably, Item 8 (TC loss + Soft-label + LSCI) shows a limited performance gap with Item 4 (InfoNCE + Soft-label + LSCI), rooted in mechanism overlap. TC loss employs an adaptive gamma strategy to adjust hard sample weights, reducing intraclass dispersion and suppressing overconfidence for calibration. Soft-label, via smoothed probability distribution, similarly promotes cross-modal feature clustering and mitigates overconfidence—aligning their core regularization objectives despite distinct implementations. This overlap compresses TC loss's improvement space when combined with soft-label, consistent with the work of [73], who showed that label smoothing alone significantly reduces ECE. When directly replacing InfoNCE with TC loss in the baseline setting (Item 1 versus Item 5),

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the ECE decreases by 4.23% (RSITMD) and 3.17% (RSICD) with negligible mR loss, demonstrating TC loss's effectiveness.

5) *Integration of All Components:* The full LSCI-TC model (No. 8), which integrates the LSCI module, soft-label, and TC loss, achieves the best overall performance, improving mR by 1.02% and reducing ECE by 0.78% compared to the variant without soft labels (No. 7). This demonstrates the complementary nature of the three components: the LSCI module ensures feature-level alignment, the soft-label provides semantic regularization, and the TC loss optimizes calibration at the output level. The combined effect validates the necessity of a holistic approach to RSITR, moving beyond isolated improvements toward joint optimization of interaction, regularization, and calibration.

2) Ablation Experiments on the Components of LSCI: Next, to further investigate the roles of each component in the LSCI module, we conducted five ablation studies, with the results presented in Table V. In this experiment, the number of heads in the cross-attention mechanism was fixed at 8, and the model was optimized using InfoNCE loss, with the top-k parameter configured to 40.

1) *No. 1:* The LSCI module solely adopts image-to-text cross-attention (ITM), which essentially leverages image-to-text similarity to enhance the representation of text features.
2) *No. 2:* The LSCI module solely adopts text-to-image cross-attention (TIM), which essentially leverages text-to-image similarity to enhance the representation of image features.
3) *No. 3:* The LSCI module integrates both text-to-image cross-attention and an MIE. Such a design is intended to bring to the fore the expressions of text-to-image similarity, thereby enhancing image features.
4) *No. 4:* The LSCI module integrates image-to-text cross-attention and multimodal interaction self-attention. This design is intended to highlight the manifestations of image-to-text similarity, thereby enhancing text features.
5) *No. 5:* The LSCI module employs both an image-to-text branch and a text-to-image branch, then integrates the features generated by these two cross-attention mechanisms through multimodal interactive self-attention.

Table V presents the results of the ablation studies on the different components of the LSCI module. In the subsequent section, we conduct a detailed comparison and analysis of these results to clarify how each component influences the overall performance of the model.

1) *No. 1 Versus No. 2 Comparison of Single Cross-Attention Mechanisms:* No. 1 (ITM-only) achieves an mR of 50.28%, outperforming No. 2 (TIM-only) which yields an mR of 49.58%. Although TIM enhances specific text-to-image metrics—such as R@5 (61.50% versus 60.38% for ITM) and R@10 (68.90% versus 67.56% for ITM)—ITM exhibits stronger overall alignment capability and more balanced performance across both retrieval directions. This result underscores the inherent bias of single cross-attention mechanisms and emphasizes the necessity of a more integrated approach, which explains why symmetric designs incorporating both ITM and TIM are adopted in works like [75].
2) *No. 1 Versus No. 3 Effect of Adding MIE to ITM:* Introducing the MIE to the ITM-only framework—resulting in configuration No. 3—improves the mR from 50.28% to 50.67%. Notably, No. 3 delivers substantial performance gains in image-to-text (I2T) retrieval across key metrics: R@1 increases from 30.94% to 32.38%, R@5 rises from 60.38% to 62.04%, and R@10 advances from 67.56% to 70.11%. These consistent improvements confirm that the MIE effectively refines and engages with ITM-enhanced features.
3) *No. 3 Versus No. 4 Comparing MIE with TIM versus ITM:* No. 4 (ITM + MIE) achieves an mR of 50.45%, slightly below No. 3 (TIM + MIE) at 50.67%, but exhibits a different performance profile: No. 4 excels notably in I2T R@1 (33.70% versus 32.38% in No. 3), indicating that TIM combined with MIE is particularly effective for high-precision image-to-text retrieval. In contrast, No. 3 maintains stronger performance in T2I R@10 (61.17% versus 60.46%), suggesting that ITM benefits longer range text-to-image recall.

Then, to investigate the influence of the initial gamma value in the TC on model performance, we evaluated the model across a range of gamma values from −2 to 2. The model was optimized using the TC in conjunction with the LSCI module.

As shown in Table VI, the model attains the highest mR value (52.36%) for standalone TC loss when the initial gamma value is set to 1.0. In terms of calibration performance, this gamma setting also yields the lowest ECE (14.93%) for TC alone, indicating superior confidence calibration in cross-modal retrieval. Notably, the proposed TC loss consistently outperforms its static focal loss (inv-focal) counterpart across nearly all gamma values in both mR and ECE. For instance, at gamma = −1.0, TC achieves an mR of 51.98% and an ECE of 16.71%, compared to 51.66% and 18.40% from static focal/inv-focal loss. This demonstrates the adaptive advantage of TC loss over static focal-based regularization, as it dynamically calibrates task-specific semantic mismatches rather than applying uniform hard-sample weighting. When expanding the comparison to other focal loss variants, we observe further evidence of TC's superiority: Sigmoid focal loss yields significantly lower retrieval performance and poorer calibration, as it is inherently designed for dense object detection tasks and lacks the ability to align cross-modal semantic representations. In contrast, dynamic-recall focal loss [76] underperforms TC and exhibits higher ECE. It only focuses on dynamic recall optimization for hard sample pairs, without incorporating task-specific calibration mechanisms to mitigate the semantic gap inherent to RSITR. In contrast, TC loss targets the domain-specific semantic mismatches of RSITR directly, balancing hard-sample mining with adaptive calibration of confidence scores—explaining its consistent superiority even over focal variants.

These findings verify that the TC loss not only boosts retrieval accuracy but also significantly enhances model

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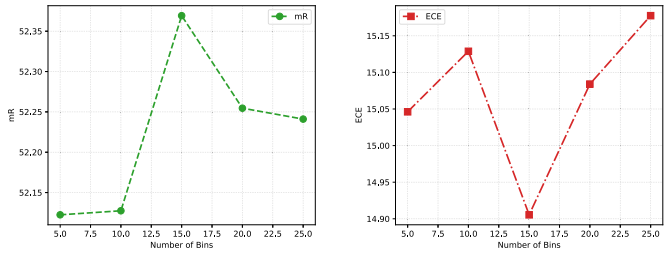


Fig. 9. Impact of different bin numbers on mR and ECE within the TC on RSITMD dataset.

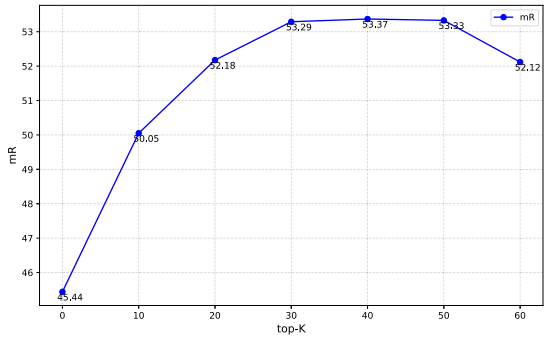


Fig. 10. Impact of the number of retrieved candidates (top-k) on mR (RSITMD dataset).

calibration, providing a more robust and reliable solution for RSITR tasks.

3) Ablation Experiments on the Components of TC: We investigate the impact of different bin counts on model performance within the TC. Specifically, we set the number of bins to range from 5 to 25, with a step size of 5.

As shown in Fig. 9, the optimal bin count for balancing mR and ECE is 15. Too few bins fail to adequately evaluate model performance, while excessive bins introduce noise and degrade performance.

4) Ablation Experiments on the Components of Retrieved Candidates: Fig. 10 illustrates the impact of varying the number of retrieved candidates (top-k) on mR performance using the RSITMD dataset. The experimental results demonstrate a clear trend where mR increases steadily as k grows from 10 to 40, reaching an optimal value of 53.37% at $k = 40$. This improvement indicates that expanding the retrieval scope enhances the model's ability to capture relevant matches within the candidate pool. However, beyond this optimal point, further increasing k to 50 and 60 yields diminishing returns and may introduce noise, as evidenced by the mR values plateauing at approximately 53.33% and 52.12%, respectively. This observed result aligns with the findings documented in [49] and [59].

V. Conclusion

In conclusion, this article proposes LSCI-TC, a framework that optimizes the contextual semantic alignment of local features and the discriminative selection of hard and easy sample pairs in RS data. The LSCI integrates cross-attention mechanisms and a multimodal interaction encoder to strengthen semantic modeling. The cross-attention mechanism enables fine-grained interactions between image and text features. Meanwhile, the MIE enhances the generation of accurate semantic representations. Together, these components advance the robustness of image–text retrieval in RS scenarios. Beyond component design, the framework's TC further complements this optimization by dynamically guiding the model to prioritize hard and easy sample pairs based on task needs. Notably, injecting fine-grained interaction is a promising direction for future research.

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